

```
# import all libraries
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

```
df=pd.read_csv("/content/archive.zip")
df
```

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin
0	892	0	3	Kelly, Mr. James	male	34.5	0	0	330911	7.8292	NaN
1	893	1	3	Wilkes, Mrs. James (Ellen Needs)	female	47.0	1	0	363272	7.0000	NaN
2	894	0	2	Myles, Mr. Thomas Francis	male	62.0	0	0	240276	9.6875	NaN
3	895	0	3	Wirz, Mr. Albert	male	27.0	0	0	315154	8.6625	NaN
4	896	1	3	Hirvonen, Mrs. Alexander (Helga E Lindqvist)	female	22.0	1	1	3101298	12.2875	NaN
...	...	...	...	...	...	...	...	...	...	...	...
413	1305	0	3	Spector, Mr. Woolf	male	NaN	0	0	A.5. 3236	8.0500	NaN
414	1306	1	1	Oliva y Ocana, Dona. Fermina	female	39.0	0	0	PC 17758	108.9000	C105
415	1307	0	3	Saether, Mr. Simon Sivertsen	male	38.5	0	0	SOTON/O.Q. 3101262	7.2500	NaN
416	1308	0	3	Ware, Mr. Frederick	male	NaN	0	0	359309	8.0500	NaN
417	1309	0	3	Peter, Master. Michael J	male	NaN	1	1	2668	22.3583	NaN

418 rows × 12 columns

Next steps: [Generate code with df](#) [New interactive sheet](#)

```
df.shape
```

```
(418, 12)
```

```
df.head() # function is used to see first few rows of a dataset
```

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
0	892	0	3	Kelly, Mr. James	male	34.5	0	0	330911	7.8292	NaN	
1	893	1	3	Wilkes, Mrs. James (Ellen Needs)	female	47.0	1	0	363272	7.0000	NaN	
2	894	0	2	Myles, Mr. Thomas Francis	male	62.0	0	0	240276	9.6875	NaN	
3	895	0	3	Wirz, Mr. Albert	male	27.0	0	0	315154	8.6625	NaN	
4	896	1	3	Hirvonen, Mrs. Alexander (Helga E Lindqvist)	female	22.0	1	1	3101298	12.2875	NaN	

Next steps: [Generate code with df](#) [New interactive sheet](#)

```
# basic info
df.info()

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 418 entries, 0 to 417
Data columns (total 12 columns):
#   Column          Non-Null Count  Dtype
---  -
0   PassengerId     418 non-null    int64
1   Survived        418 non-null    int64
2   Pclass          418 non-null    int64
3   Name            418 non-null    object
4   Sex             418 non-null    object
5   Age            332 non-null    float64
6   SibSp           418 non-null    int64
7   Parch           418 non-null    int64
8   Ticket          418 non-null    object
9   Fare           417 non-null    float64
10  Cabin           91 non-null     object
11  Embarked        418 non-null    object
dtypes: float64(2), int64(5), object(5)
memory usage: 39.3+ KB
```

df.describe()

	PassengerId	Survived	Pclass	Age	SibSp	Parch	Fare	
count	418.000000	418.000000	418.000000	332.000000	418.000000	418.000000	417.000000	
mean	1100.500000	0.363636	2.265550	30.272590	0.447368	0.392344	35.627188	
std	120.810458	0.481622	0.841838	14.181209	0.896760	0.981429	55.907576	
min	892.000000	0.000000	1.000000	0.170000	0.000000	0.000000	0.000000	
25%	996.250000	0.000000	1.000000	21.000000	0.000000	0.000000	7.895800	
50%	1100.500000	0.000000	3.000000	27.000000	0.000000	0.000000	14.454200	
75%	1204.750000	1.000000	3.000000	39.000000	1.000000	0.000000	31.500000	
max	1309.000000	1.000000	3.000000	76.000000	8.000000	9.000000	512.329200	

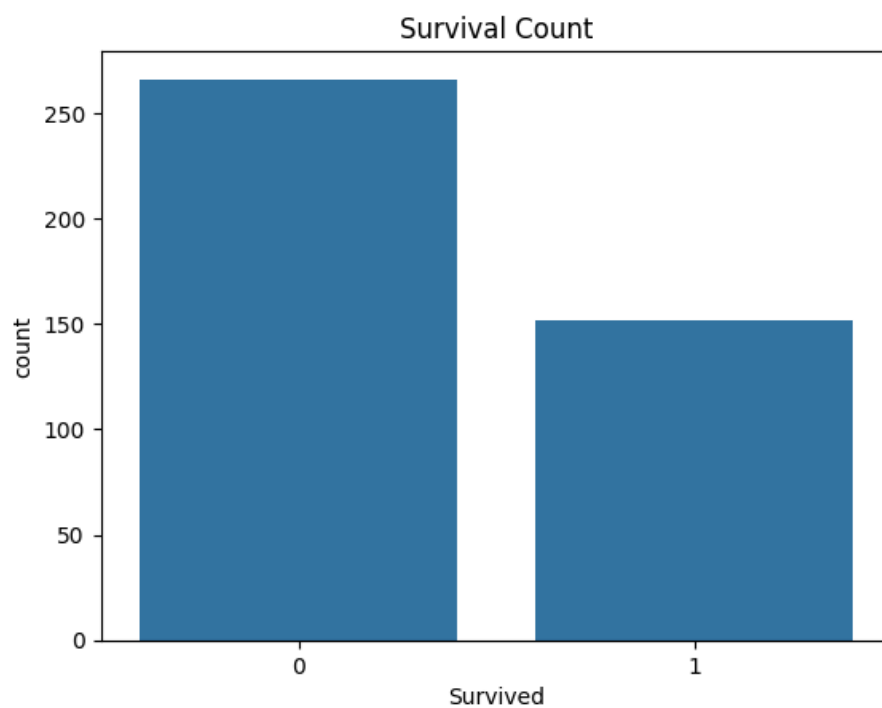
```
# missing values check
df.isnull().sum()
```

	0
PassengerId	0
Survived	0
Pclass	0
Name	0
Sex	0
Age	86
SibSp	0
Parch	0
Ticket	0
Fare	1
Cabin	327
Embarked	0

dtype: int64

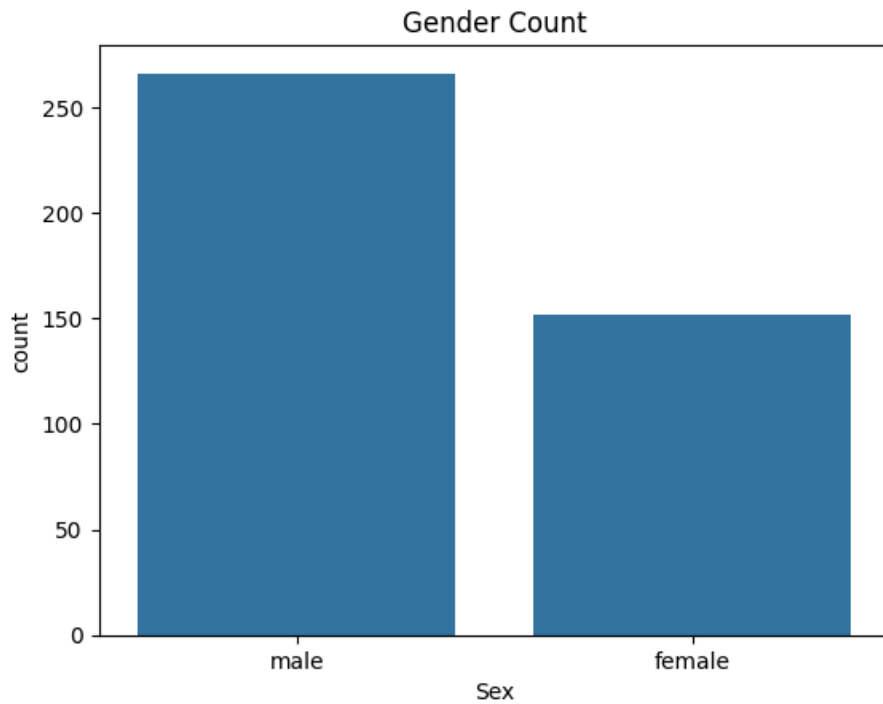
```
# clean missing values
df["Age"]=df["Age"].fillna(df["Age"].median())
df["Embarked"]=df["Embarked"].fillna(df["Embarked"].mode()[0])
```

```
# EDA Visualizations
# Survival count
sns.countplot(x='Survived', data=df)
plt.title("Survival Count")
plt.show()
```



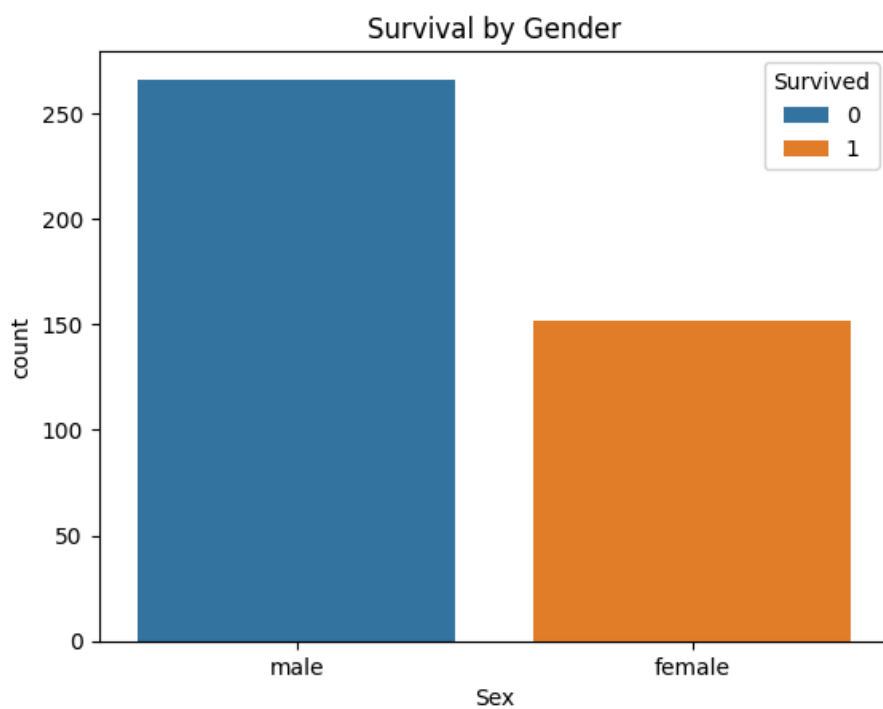
More people did not survive compared to those who survived. Survival rate is low

```
# Gender Distribution
sns.countplot(x='Sex', data=df)
plt.title("Gender Count")
plt.show()
```



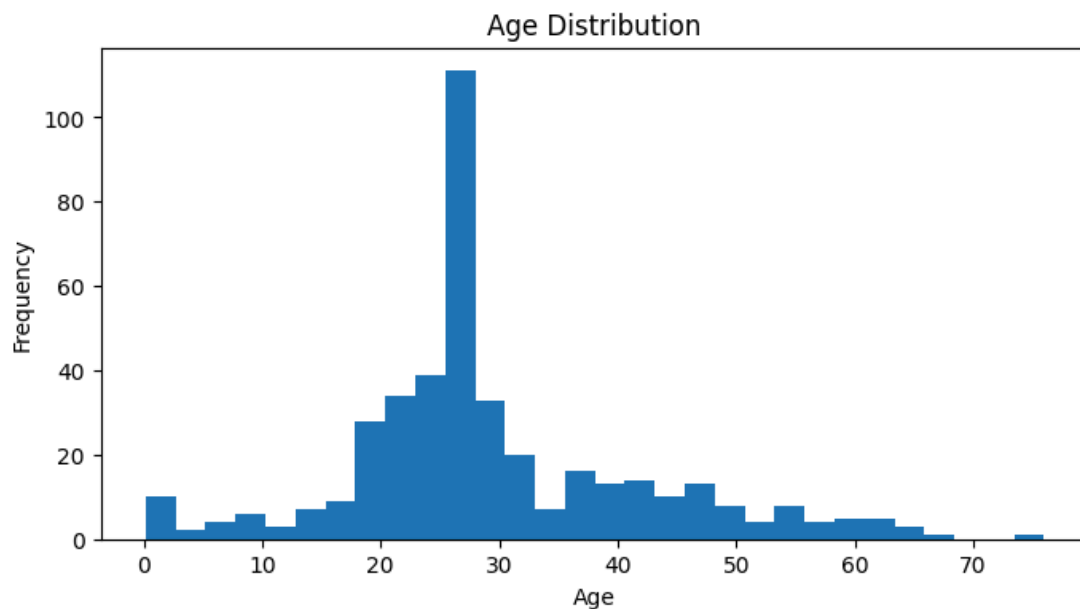
More male passengers than females on the ship

```
#Survival by Gender
sns.countplot(x='Sex', hue='Survived', data=df)
plt.title("Survival by Gender")
plt.show()
```



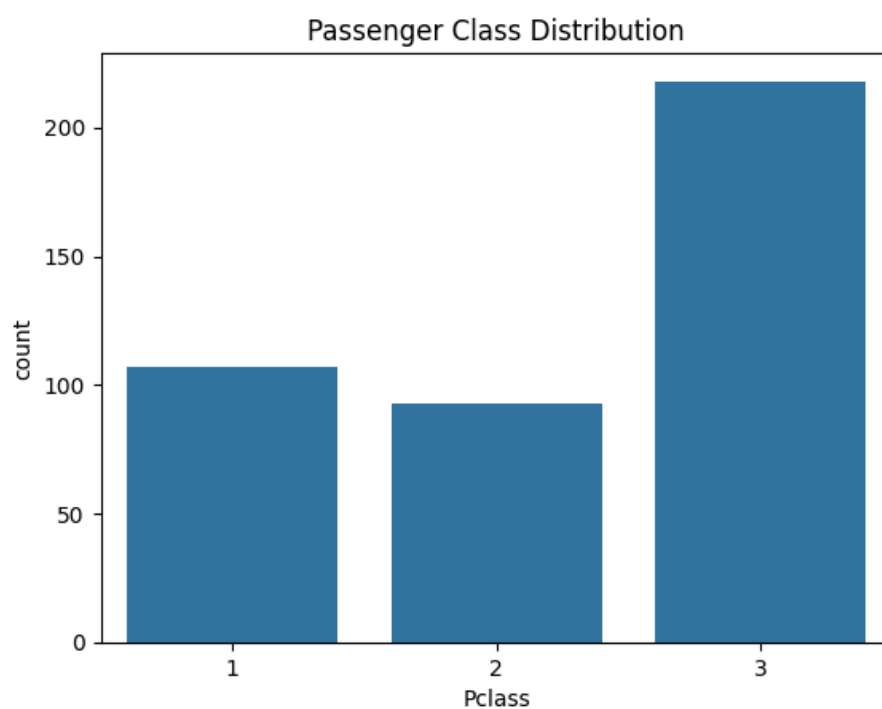
• Females survived more than males.

```
#Age Distribution
df['Age'].plot(kind='hist', bins=30, figsize=(8,4))
plt.title("Age Distribution")
plt.xlabel("Age")
plt.show()
# Majority travelled in 3rd Class (cheapest). • Few people in 1st Class (rich passengers).
```

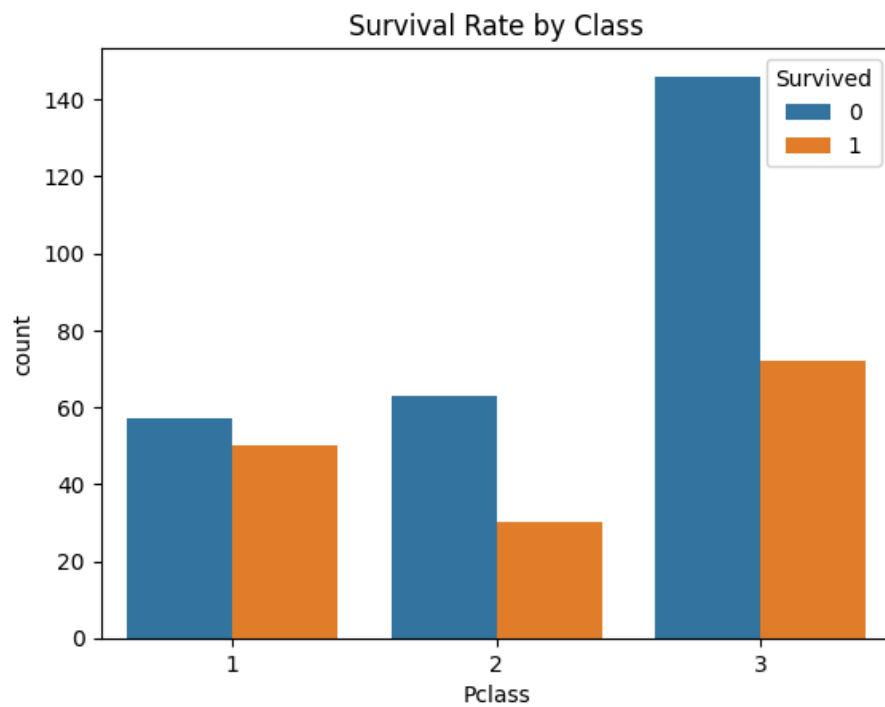


Most passengers were between 20–40 years. • Very few older passengers (above 60).

```
# Passenger Class Count
sns.countplot(x='Pclass', data=df)
plt.title("Passenger Class Distribution")
plt.show()
```

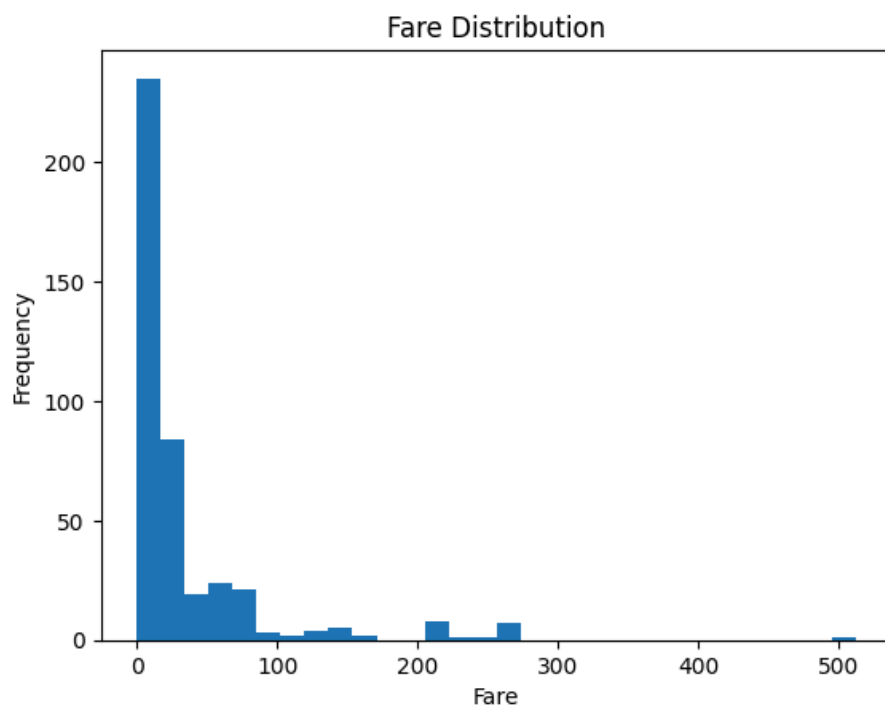


```
#Survival by Passenger Class
sns.countplot(x='Pclass', hue='Survived', data=df)
plt.title("Survival Rate by Class")
plt.show()
```



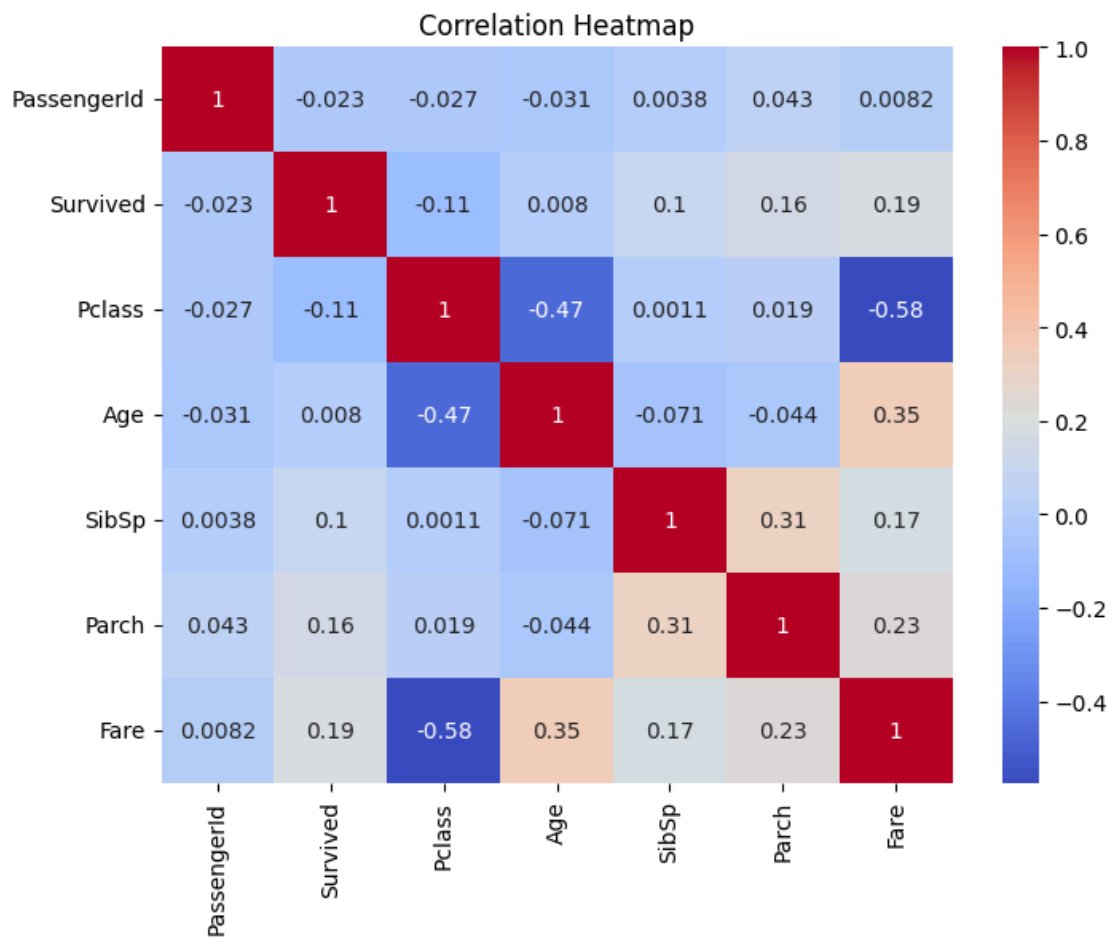
- 1st class passengers survived more than 2nd and 3rd class.
- 3rd class survival is very low.

```
#Fare distribution
df['Fare'].plot(kind='hist', bins=30)
plt.title("Fare Distribution")
plt.xlabel("Fare")
plt.show()
```

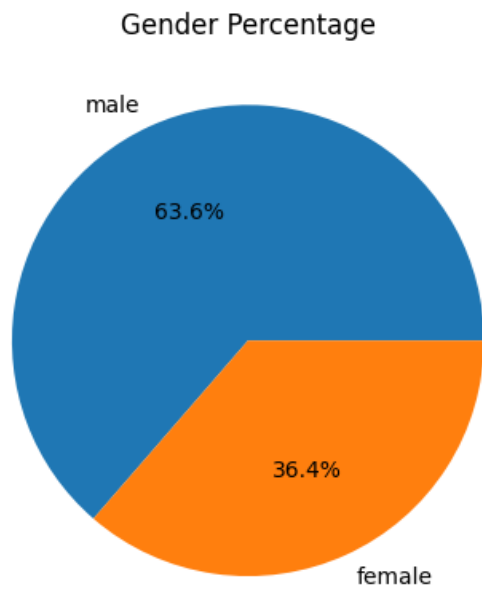


- Most fares are low (cheaper tickets).
- Few very expensive tickets → mostly 1st class.

```
#Correlation Heatmap
numeric_df = df.select_dtypes(include=['int64', 'float64'])
plt.figure(figsize=(8,6))
sns.heatmap(numeric_df.corr(), annot=True, cmap='coolwarm')
plt.title("Correlation Heatmap")
plt.show()
```



```
df['Sex'].value_counts().plot(kind='pie', autopct='%1.1f%%')  
plt.title("Gender Percentage")  
plt.ylabel("")  
plt.show()
```



```
sns.pairplot(numeric_df)  
plt.show()
```

